Week 6

1. Cluster analysis is primarily a:

A) Supervised learning technique
B) Reinforcement learning technique
C) Unsupervised learning technique
D) Deep learning technique

Answer: C

Explanation

A. Supervised learning technique (Incorrect): Supervised learning requires labeled training data, whereas clustering works without labels.

B. Reinforcement learning technique (Incorrect): Reinforcement learning learns through rewards and penalties from interactions with an environment, which is unrelated to clustering.

C. Unsupervised learning technique (Correct): Cluster analysis groups similar data objects into clusters without using predefined class labels, making it an unsupervised learning method.

D. Deep learning technique (Incorrect): Deep learning is a separate class of machine learning models based on neural networks; clustering itself is not inherently a deep learning method.

2. What is the main goal of cluster analysis?

A) Predict future values
B) Organize similar items into groups
C) Classify labeled data
D) Reduce storage space

Answer: B

Explanation

A. Predict future values (Incorrect): Predicting future values is typically the goal of regression and forecasting techniques, not clustering.

B. Organize similar items into groups (Correct): The primary objective of clustering is to group data objects so that items within the same cluster are more similar to each other than to those in other clusters.

C. Classify labeled data (Incorrect): Classification is a supervised learning task that requires predefined class labels, whereas clustering does not.

D. Reduce storage space (Incorrect): Although clustering may help in data summarization, its main purpose is discovering natural groupings in data, not reducing storage requirements.

3. Which similarity measure calculates straight-line distance between two points?

A) Manhattan Distance
B) Cosine Similarity
C) Euclidean Distance
D) Hamming Distance

Answer: C

Explanation

A. Manhattan Distance (Incorrect): Manhattan Distance measures distance as the sum of absolute differences along dimensions, like moving along city blocks rather than a straight line.

B. Cosine Similarity (Incorrect): Cosine Similarity measures the angle between two vectors and indicates similarity in direction, not physical distance.

C. Euclidean Distance (Correct): Euclidean Distance measures the shortest (straight-line) distance between two points in Euclidean space and is widely used in clustering algorithms such as K-Means.

D. Hamming Distance (Incorrect): Hamming Distance counts the number of differing positions between two strings or binary vectors, not the straight-line distance between points.

4. Which similarity measure is based on the angle between vectors?

A) Euclidean Distance
B) Manhattan Distance
C) Cosine Similarity
D) Jaccard Distance

Answer: C

Explanation

A. Euclidean Distance (Incorrect): Euclidean Distance measures the straight-line distance between two points, not the angle between vectors.

B. Manhattan Distance (Incorrect): Manhattan Distance calculates the sum of absolute differences across dimensions and does not consider vector angles.

C. Cosine Similarity (Correct): Cosine Similarity measures the cosine of the angle between two vectors, indicating how similar their directions are regardless of magnitude.

D. Jaccard Distance (Incorrect): Jaccard Distance measures dissimilarity between sets based on intersection and union, not the angle between vectors.

5. Which is a common application of cluster analysis?

A) Customer segmentation
B) Sorting algorithms
C) Compiler optimization
D) CPU scheduling

Answer: A

Explanation

A. Customer segmentation (Correct): Cluster analysis is widely used to group customers with similar behaviors, preferences, or purchasing patterns for targeted marketing and business decision-making.

B. Sorting algorithms (Incorrect): Sorting algorithms arrange data in a specific order, whereas clustering groups similar data objects.

C. Compiler optimization (Incorrect): Compiler optimization improves program execution efficiency and is not a typical application of cluster analysis.

D. CPU scheduling (Incorrect): CPU scheduling determines the execution order of processes in an operating system and does not involve clustering techniques.

6.Which algorithm is most commonly associated with cluster analysis?

A) Decision Tree
B) Naïve Bayes
C) K-Means
D) Logistic Regression

Answer: C

Explanation

A. Decision Tree (Incorrect): Decision Trees are supervised learning algorithms used for classification and regression, not clustering.

B. Naïve Bayes (Incorrect): Naïve Bayes is a supervised probabilistic classification algorithm that requires labeled data.

C. K-Means (Correct): K-Means is one of the most widely used clustering algorithms that partitions data into K clusters based on similarity and distance measures.

D. Logistic Regression (Incorrect): Logistic Regression is a supervised learning technique used for classification tasks, not for grouping unlabeled data into clusters.

7. What is the first step of the K-Means algorithm?

A) Compute accuracy
B) Select initial centroids
C) Normalize output
D) Remove outliers

Answer: B

Explanation

A. Compute accuracy (Incorrect): Accuracy is generally used in supervised learning and is not the first step of K-Means clustering.

B. Select initial centroids (Correct): K-Means begins by choosing K initial centroids, which act as the starting cluster centers for assigning data points.

C. Normalize output (Incorrect): Data normalization may be performed as a preprocessing step, but it is not an inherent first step of the K-Means algorithm itself.

D. Remove outliers (Incorrect): Outlier removal is optional data preprocessing and not a mandatory first step in K-Means.

8. Which metric is commonly used to evaluate K-Means clustering quality?

A) Accuracy
B) Recall
C) WSSE
D) Precision

Answer: C

Explanation

A. Accuracy (Incorrect): Accuracy is a classification metric that requires labeled data, whereas K-Means is an unsupervised learning algorithm.

B. Recall (Incorrect): Recall measures how many relevant instances are correctly identified in classification problems and is not used for evaluating clustering quality.

C. WSSE (Correct): WSSE (Within-Set Sum of Squared Errors) measures the total squared distance between data points and their assigned cluster centroids. Lower WSSE generally indicates better cluster compactness.

D. Precision (Incorrect): Precision is another classification metric based on labeled data and is not a standard measure for K-Means clustering.

9. If WSSE decreases, it generally indicates:

A) Better cluster compactness
B) Worse clustering
C) More classification errors
D) Data loss

Answer: A

Explanation

A. Better cluster compactness (Correct): A lower WSSE (Within-Set Sum of Squared Errors) means data points are closer to their cluster centroids, indicating more compact and cohesive clusters.

B. Worse clustering (Incorrect): A decrease in WSSE generally suggests improved clustering quality, not worse clustering.

C. More classification errors (Incorrect): WSSE is a clustering metric and is unrelated to classification errors.

D. Data loss (Incorrect): WSSE measures clustering quality and has nothing to do with loss of data.

10. K-Means stops iterating when:

A) No changes occur in centroids
B) Accuracy becomes 100%
C) Dataset becomes empty
D) Number of features decreases

Answer: A

Explanation

A. No changes occur in centroids (Correct): K-Means converges and stops when cluster centroids no longer change (or change only negligibly) between iterations, indicating stable clusters.

B. Accuracy becomes 100% (Incorrect): K-Means is an unsupervised algorithm and does not use accuracy as a stopping criterion.

C. Dataset becomes empty (Incorrect): The dataset remains unchanged during clustering; K-Means does not stop because the dataset becomes empty.

D. Number of features decreases (Incorrect): The number of features is fixed for a dataset and is not a condition for terminating the K-Means algorithm.